# A MAP TO A SMOOTH S-SCHEME FROM A PERFECT S-SCHEME IS PERFECT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 17833-17844 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17846-17866, the complete author proof (19 body lines).

*Editorial note.* Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Permanence member of the chapter's perfect-morphism series (with L16092 descent and L16093 smooth-diagonal characterization, both collected); the sibling regular-immersion-perfect lemma it cites is itself a top-30 candidate in the remaining queue and will close that internal edge next wave.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Chapter lemmas flat-base-change-perfect, regular-immersion-perfect, composition-perfect; Morphisms-chapter smooth-is-flat and base-change-smooth; Divisors, lemma section-smooth-regular-immersion.

*Source reference.* Original lemma 067R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17866–17866. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0689: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17760–17762. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0688: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17744–17746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 068C: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17814–17817. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01VB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17866–17866. Cited by the retained statement or

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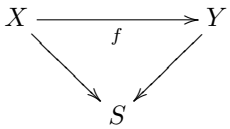
proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01VF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17866–17866. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let*




*be a commutative diagram of morphisms of schemes. Assume $Y \to S$ smooth and $X \to S$ perfect. Then $f : X \to Y$ is perfect.*

*Proof.* We can factor $f$ as the composition

$$X \longrightarrow X \times_S Y \longrightarrow Y$$

where the first morphism is the map $i = (1, f)$ and the second morphism is the projection. Since $Y \to S$ is flat, see Morphisms, Lemma 01VF, we see that $X \times_S Y \to Y$ is perfect by Lemma 0688. As $Y \to S$ is smooth, also $X \times_S Y \to X$ is smooth, see Morphisms, Lemma 01VB. Hence $i$ is a section of a smooth morphism, therefore $i$ is a regular immersion, see Divisors, Lemma 067R. This implies that $i$ is perfect, see Lemma 068C. We conclude that $f$ is perfect because the composition of perfect morphisms is perfect, see Lemma 0689. □